

Deep Learning for Structured Data – Exercise Sheet 1

Submit the solution in a single PDF in the Moodle until 24.04.2026, 08:30. The corresponding exercise session will take place on the same day at 08:30.

Make yourself familiar with PyTorch, e.g., by following the tutorial at <https://docs.pytorch.org/tutorials/beginner/basics/intro.html>.

Problem 1: (5 Pt.)

Implement a feed-forward neural network in PyTorch.

1. Construct a synthetic regression dataset with a train and test dataset, where a one-layer network does not show good performance on the test dataset, while a network with more layers shows good performance.
2. Can you propose a regularization term that reduces the performance gap between the test and train datasets for the larger network? Show empirically that your regularizer works.

Problem 2: (5 Pt.)

Derive a hypothesis class $\mathcal{H}$ and a data distribution $\mathbb{P}$ such that there exists a hypothesis $h \in \mathcal{H}$ for which the training error is zero, but the expected test error is no better than random guessing.

1. Define a finite hypothesis class $\mathcal{H}$ with at least two hypotheses and describe a data distribution $\mathbb{P}$ on $(\mathcal{X}, \mathcal{Y})$.
2. Construct a small training set drawn from $\mathbb{P}$ where one hypothesis h_{overfit} fits the training data perfectly, achieving zero training error.
3. Explain why, under the chosen $\mathbb{P}$, this hypothesis does no better than random guessing on unseen test data.

Problem 3: (5 Pt.)

Assume that $0 \leq \ell \leq 1$ and that the hypothesis class $\mathcal{H}$ is finite. Let

$$h^* \in \arg \min_{h \in \mathcal{H}} R(h), \quad \hat{h}_n \in \arg \min_{h \in \mathcal{H}} \hat{R}_n(h).$$

Suppose that with probability at least $1 - \delta$,

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_n(h)| \leq \varepsilon, \quad \text{where } \varepsilon := \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2n}}.$$

Show that on this event,

$$R(\hat{h}_n) \leq R(h^*) + 2\varepsilon.$$